

PROJECT OVERVIEW

AIR PIANO

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PROBLEM STATEMENT AND SUCCESS CRITERIA

Project Description

- ▶ Our project is the **Air Piano**, a contact-less digital instrument. Users play notes by making motions with their hand in front of a break beam sensor. The pitch of each note is determined by the distance from the device to their hand as measured by an ultrasonic distance sensor. Like a classical piano, pitches are quantized based on the standard 12-TET tuning system. To increase playability, we chose to limit the device to eight notes, forming a C major scale. Finally, the device includes a real-time LED display indicating which pitch is currently to be played based on the distance sensor.

Success Criteria

1. The device plays a single note for a short time (configurable) on the buzzer only when a hand moves downwards in front of it.
2. The pitch of the note is lower when the hand is closer in horizontal distance to the device and higher when it is further away. The pitches played by the device will range from C4 to C5.
3. The LED display on the device correctly displays the note that would be played should the user lower the hand closest to the device at that moment.
4. The timing requirements for the device are met.

HARDWARE

SBC Choice

- ▶ Beaglebone Black

Electronics Kit Components

- ▶ Breadboard
- ▶ Jumper wires
- ▶ Passive buzzer
- ▶ LEDs

Add-On Sensors

- ▶ IR break beam sensor
- ▶ HC-SR04 ultrasonic distance sensor

REQUIREMENTS

Measurable Timing Requirements

1. On note-down event as measured by the break beam sensor, audio output must begin with 10ms worst-case.
 2. On note-change event as measured by the distance sensor, the LED display must begin updating within 100ms worst-case.
 3. Break beam sensor is to be sampled at 100Hz with jitter ± 1 ms.
 4. Distance sensor is to be sampled at 30Hz with jitter ± 1 ms.
- *See Wessel and Wright, 2002*

REQUIREMENTS

Periodic Tasks

1. Take break beam measurements every 10ms (100Hz)
2. Take distance measurements every 33ms (30Hz)
3. Update LED display every 50ms (20Hz)
4. Write logs every 100 ms (10 Hz)

Asynchronous Task

- Upon beam break detection, play current note within 10ms.

ARCHITECTURE

Task Architecture

Task Name	Priority	Responsibilities
Beam Sensor	90	Detects break beam being broken and plays the note determined by distance when triggered.
Distance Sensor	70	Measures distance from sensor and calculates corresponding note.
Display Updater	60	Continuously displays the current note indicated by the distance sensor.
Console	40	Reads special commands from console input.
Logger	30	Outputs both real-time updates to console output as well as logs to file.

Task Communication

- ▶ The Distance Sensor will continuously update a shared value with the current note that corresponds to the distance measurement. This allows the Display Updater and the Beam Sensor to read this shared value to quickly play or display the note. The data will be protected by a lock.
- ▶ Tasks will communicate with the Logger via buffers protected by a lock. The logger will only hold onto the lock while copying data to its own memory.
- ▶ The safe state will be protected by a lock, and different tasks will update it based on their specific failure-detection criteria.

ARCHITECTURE

Safe-State Behavior

- ▶ The safe-state behavior of the Air Piano will turn off all LEDs and output no sound. This will signal that something is wrong and prevent annoyance from an indefinitely held note.
- ▶ If more than 10 beam breaks are detected within one second, the safe state will be entered, as this indicates a possible sensor or logical malfunction.
- ▶ If the distance sensor detects a value outside of the playable note range, no audio will be produced, but the device will not enter the safe state. If the distance sensor value is clearly outside of the sensor's own range, the safe state will be entered.
- ▶ To exit the safe state, an admin will enter a command on the console.
 - Admins may also manually enter the safe state via console command.

TESTS AND MEASUREMENTS

Timing Data

1. Capture timestamps for both beam breaks and audio playback to ensure deadlines are met. Log to file for review and testing.
2. Also record captured distance with notes played to verify sensor logic and overall functionality. Store in file for review and testing.

Boundary and Load Cases

1. Test with high sensor load (8 beam break events per second) to verify deadlines are met and functionality is retained.
2. Test with multiple objects around the sensor such as people walking in the background to ensure functionality isn't impeded by unwanted interference (no spurious note playback events).
3. Test with high CPU load by running other processes in the background.

TESTS AND MEASUREMENTS

Base Timing Measurements

- ▶ While most deadlines were met, our measurements showed that the system occasionally missed deadlines, even for the highest priority tasks.
- ▶ Our measurements indicate very low latency to output the note once detected, with worst case latency less than 1ms. Likewise, the latency for the display was always less than 100ms.

Load Conditions

- ▶ By simulating 8 beam break events per second, we confirmed that the system functioned as normal.
- ▶ From ad-hoc experiments, we determined that noise from the external environment can affect the operation of the device (e.g. objects near the distance sensor).
- ▶ At 100% CPU load, we found our timing data to be comparable to the baseline.

REFERENCES I



Wessel, D., & Wright, M. (2002). **Problems and prospects for intimate musical control of computers.**
Computer Music Journal, 26(3), 11–22.